

Route-F P54: Common Renormalization, a Radial Stability Obstruction and C_{HN} Threshold Closure

P54PQ-v2 four-dimensional mainline

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Abstract

We verify the finite-parameter conversion linking the existing fixed-VEV and minimal-subtraction descriptions of the same bare scalar action. Deleting the large finite counterterm while retaining the original renormalized parameters is not such a conversion. We calculate the previously untested three-dimensional physical radial bosonic Hessian. It has a negative eigenvalue in the declared one-loop hard-potential truncation. An analytical bound proves that no choice of the three Majorana masses can remove this negative direction in that truncation at the frozen scalar point. This is a conditional benchmark obstruction, not an all-orders or gauge-independent exclusion of the model. Separately, we calculate the linear- C_{HN} finite Yukawa vertex and associated terminal Weinberg and Higgs-mass contributions. At partial sterile thresholds the field-dependent Schur map generates a dimension-six $LNH(H^\dagger H)$ operator. Its mass-dependent mixing explains a precise residual left by a dimension-five-only sequential treatment when retained Majorana mass effects are not additionally expanded away.

1 Scope and fixed input

The action remains P54PQ-v2: $3\,16_F + 54_{H,\mathbb{R}} + 126_{H,\mathbb{C}} + 10_{H,\mathbb{C}} + 1_{H,\mathbb{C}}$, with the two complex symmetric family matrices $h_{\text{raw}}, f_{\text{raw}}$. No phenomenological fit, new family parameter, lattice scan or replacement scalar point is introduced. The historical broken tree vacuum and upper tree PS saddle are respectively

$$r_B = (w, \sigma, v_s) = (1, 0.1265, 0.25) \omega, \quad (1)$$

$$r_U = (1.0008103059, 0, 0.2542776971) \omega. \quad (2)$$

Here the notation $w = 1 \omega$ uses the same historical reference scale as the earlier files. Both points use the same scalar parameters. We retain DR/ $\overline{\text{MS}}$, background-field Landau gauge, and the existing finite invariant fixed-VEV mass counterterms at $\mu_0 = 0.5626028899853225 \omega$. Write $\mathcal{L} = 16\pi^2$.

“Hard potential” means the previously declared trace over 290 positive scalar and 33 massive vector eigenvalues at r_B , with the 38 scalar and 12 vector soft directions excluded and mixed hard/soft divided differences retained. It is not the complete soft-IR-resummed effective action. Its Hessian is not a pole mass. This scope applies to every stability statement below.

2 Finite counterterms cannot be removed at fixed numerical input

Let $p = (\mu^2, \nu^2, \mu_s^2)$ be the three dependent mass parameters. The canonical real-field Hessian of their finite counterterm is

$$\text{H}_{\text{CT}} = \text{diag} \left(-\Delta \mu^2 I_{54}, -\frac{1}{2} \Delta \nu^2 I_{252}, 0_{20}, -\Delta \mu_s^2 I_2 \right). \quad (3)$$

The factor $1/2$ for the complex 126 is required by the action’s normalization and is independently consistent with the radial and phase Ward tests. It is not a missing factor causing the large correction.

Introduce a formal loop-counting parameter ℓ . Holding the other counterterm conventions fixed, equality of the bare parameters gives

$$p_0 = p_F + \delta p_{\text{pole}} + \ell \Delta p = p_{\overline{\text{MS}}} + \delta p_{\text{pole}} + O(\ell^2), \quad p_{\overline{\text{MS}}} = p_F + \ell \Delta p + O(\ell^2). \quad (4)$$

Because these mass parameters enter V_0 linearly, at fixed fields

$$V_0(X; p_F + \ell \Delta p) = V_0(X; p_F) + \frac{\ell}{2} X^T H_{\text{CT}} X \quad (5)$$

is an exact polynomial identity. Replacing p_F by $p_{\overline{\text{MS}}}$ inside a one-loop diagram changes it only at formal two-loop order; that order statement does not guarantee a numerically small remainder at $\ell = 1$.

Parameter / ω^2	p_F	Δp	$p_F + \Delta p$
μ^2	1.4095253	-0.0563307	1.3531946
ν^2	0.3819020	-5.4697441	-5.0878421
μ_s^2	0.1719203	-0.0342377	0.1376826

Thus using zero finite counterterms with the old p_F does not describe the same bare point. Conversely, a literal resummation of the large converted mass shift is not automatically a controlled implementation of the formal scheme change. The finite scheme and VEV issues are distinct from gauge invariance of observables [1]; the present fixed-Landau calculation does not establish an FJ or GIVS gauge-independent parameter construction.

2.1 Vacuum transport and the common tadpole condition

On the radial slice, the canonical metric and mixed parameter derivative are

$$G_r = \text{diag}(12/5, 2, 1), \quad A_{ri} = \frac{\partial^2 V_0}{\partial r_r \partial p_i} = -\text{diag}(12w/5, \sigma, v_s). \quad (6)$$

Let $H_r = \partial_r^2 V_0$ and $t = \partial_r V_{1,h}$ at r_B . The fixed-VEV condition is exactly

$$t + A \Delta p = 0. \quad (7)$$

In the converted MS description the tree stationary point shifts by

$$\delta r_{\text{tree}}^{\overline{\text{MS}}} = -H_r^{-1} A \Delta p = H_r^{-1} t, \quad \delta r_{\text{explicit loop}} = -H_r^{-1} t. \quad (8)$$

Their sum vanishes at first order; neither contribution may be discarded. The calculated tree conversion coefficient is

$$\delta r_{\text{tree}}^{\overline{\text{MS}}}/\omega = (0.306642, -25.428874, 1.660472). \quad (9)$$

This coefficient is far too large to assert a small shift at $\ell = 1$. Three tiny formal values $\ell = 10^{-5}, 5 \times 10^{-6}, 2.5 \times 10^{-6}$ verify the stationary-point identity and quadratic remainder. They are local scheme-identity tests, not a new physical vacuum scan.

At the upper PS saddle, only the two singlet radii w, v_s carry tadpoles. Let H_U be their tree Hessian. Then the previously calculated hard-plus-CT response separates unambiguously as

$$\delta r_U = -H_U^{-1}(t_{U,h} + t_{U,\text{CT}}) = \delta r_{U,\text{loop}} + \delta r_{U,\text{scheme}}. \quad (10)$$

The same mass counterterms apply at both backgrounds. Their upper $\sigma = 0$ tadpole cannot determine a second independent $\Delta \nu^2$. This is the scalar-valley version of the tadpole-insertion issue in matching [2]; that reference excludes heavy gauge bosons and is not being used as a complete gauge-matching formula.

2.2 Wilson matching must transform with the parameters

For an EFT relation $q = f(p) + \ell\Delta(p)$ and finite definitions $p' = p + \ell a$, $q' = q + \ell b$, expansion of the same relation yields

$$\Delta' = \Delta + b - Df a. \quad (11)$$

The finite UV and EFT parameter cards therefore both matter. For the upper scalar exchange $C_0 = J^T D^{-1} J$, let

$$D_p = H_{CT,h} + \sum_{r=w,v_s} \delta r_{U,\text{scheme}}^r \partial_r D_0. \quad (12)$$

The original upper eigenbasis is held fixed as a field-coordinate chart, so its Yukawa source tensor J does not independently change. The tree Wilson coefficient shifts by

$$\delta_p C_0 = -J^T D_0^{-1} D_p D_0^{-1} J, \quad \Delta C_{\overline{\text{MS}}} = \Delta C_F - \delta_p C_0. \quad (13)$$

Their sum is the old full first-order response. Exact-resolvent finite differences test the tree derivative on the existing two synthetic source cases. These identities close the stated common bosonic scheme algebra, not the missing Wilson boxes or gauge/ghost/Yukawa diagrams.

3 Previously untested physical radial block

3.1 Actual action derivatives and an independent spectral check

Let R embed the three radial coordinates in the 328 real fields and $H(X) = V_0''(X)$. The quartic action makes H quadratic. Using cached first and diagonal second derivatives, three new mixed Hessian evaluations determine all $H_{,rs}$ exactly as a polynomial identity. No spectrum formula is fitted.

Let $F_{c,h}$ equal $x^2[\log(x/\mu_0^2) - c]$ in neighborhoods of the hard eigenvalues and zero near the excluded soft eigenvalues. The spectral gap makes this a smooth local spectral function. Define

$$F_{c,h}^{[1]}(x, y) = \begin{cases} [F'_{c,h}(x) - F'_{c,h}(y)]/(x - y), & x \neq y, \\ F''_{c,h}(x), & x = y. \end{cases} \quad (14)$$

Both sums below include hard and soft indices; a mixed divided difference need not vanish. In the common eigenbasis,

$$\partial_r \partial_s \text{tr} F_{c,h}(H) = \sum_i F'_{c,h}(m_i^2)(H_{,rs})_{ii} + \sum_{ij} F_{c,h}^{[1]}(m_i^2, m_j^2)(H_{,r})_{ij}(H_{,s})_{ji}. \quad (15)$$

Scalar and vector multiplicities are 1 and 3, with $c = 3/2$ and $5/6$, divided by $64\pi^2$. Direct finite differences of the actual spectral potential at two step sizes independently verify the resulting radial Hessian, not just its symmetry.

The tree and corrected bosonic generalized squared masses solve $H_r v = m^2 G_r v$ and are

$$\text{spec}(G_r^{-1} H_{r,0})/\omega^2 = (0.0135205, 0.0997157, 2.8291659), \quad (16)$$

$$\text{spec}(G_r^{-1} H_{r,B})/\omega^2 = (-0.7584480, 0.0941189, 3.0931051), \quad (17)$$

$$H_{r,B} = H_{r,0} + \Pi_{r,S} + \Pi_{r,V} + R^T H_{CT} R. \quad (18)$$

The actual matrix is approximately

$$H_{r,B}/\omega^2 = \begin{pmatrix} 7.421952 & -0.056046 & 0.057186 \\ -0.056046 & -1.516518 & -0.006028 \\ 0.057186 & -0.006028 & 0.094555 \end{pmatrix}. \quad (19)$$

Its negative direction is orthogonal to both the gauge orbit and the PQ phase. It is not an unremoved gauge zero mode. The relative insertion $H_{r,0}^{-1/2}(H_{r,B} - H_{r,0})H_{r,0}^{-1/2}$ has a smallest eigenvalue about -56.8 . The earlier doublet-only positivity check was never a whole-vacuum stability certificate.

The large $\Delta\nu^2$ is dominated by actual high-multiplicity scalar clusters. Four leading degenerate groups have multiplicities 50, 44, 44, 36 and contribute approximately -1.1353 , -1.0839 , -1.0669 , and -0.7821 to $\Delta\nu^2/\omega^2$. These are basis-invariant spectral traces. The complete group inventory is in the JSON report; there is no evidence of a decimal-level kernel error explaining the instability.

4 An all-family bound within the declared hard truncation

At the broken radial background the actual Yukawa tensors give one massive spinor direction per family, namely the Majorana neutrino. The three Takagi masses are $M_i = \sigma f_i$, $f_i \geq 0$. There is no h_{raw} radial fermion mass, and w, v_s do not enter these masses. The scalar doublet retuning also has zero radial Hessian. These statements are checked directly in the existing Clifford dictionary.

Theorem 1 (Fixed-point Majorana bound). *Fix the scalar parameters, r_B , μ_0 , the hard-potential truncation specified above, and the common fixed-VEV prescription. For any three Majorana Takagi masses, including their zero limits, adding their one-loop potential and associated finite tadpole counterterm leaves a negative radial direction. This statement does not include soft-IR completion, higher loops, pole matching, or a changed scalar point.*

Proof. The fermion potential is

$$V_F = -\frac{1}{32\pi^2} \sum_{i=1}^3 M_i^4 \left(\log \frac{M_i^2}{\mu_0^2} - \frac{3}{2} \right). \quad (20)$$

Differentiate with respect to σ at fixed f_i, μ_0 . The finite fixed-VEV counterterm has $\Delta\nu_F^2 = (\partial_\sigma V_F)/\sigma$. Consequently its only nonzero radial curvature is

$$F_{\sigma\sigma} = \partial_\sigma^2 V_F - \frac{\partial_\sigma V_F}{\sigma} \quad (21)$$

$$= -\frac{4}{\mathcal{L}\sigma^2} \sum_{i=1}^3 M_i^4 \log \frac{M_i^2}{\mu_0^2}. \quad (22)$$

For $x \geq 0$, the continuous function $-x^2 \log(x/\mu_0^2)$ has its global maximum $\mu_0^4/(2e)$ at $x = \mu_0^2 e^{-1/2}$. Thus

$$F_{\sigma\sigma} \leq F_{\max} = \frac{6\mu_0^4}{e\mathcal{L}\sigma^2} = 0.08751124\omega^2. \quad (23)$$

This bound allows arbitrarily large f_i ; no perturbativity cutoff or flavor fit is needed for the inequality. In the Loewner order,

$$H_{r,B} + H_{r,F} \preceq H_{r,B} + F_{\max} e_\sigma e_\sigma^T. \quad (24)$$

Already the fixed normalized direction $q = e_\sigma/\sqrt{2}$ obeys

$$q^T G_r q = 1, \quad q^T (H_{r,B} + H_{r,F}) q \leq \frac{-1.51651817 + 0.08751124}{2} \omega^2 < 0. \quad (25)$$

An optimized generalized-eigenvector witness for the upper-bound matrix gives the slightly stronger value $-0.7146953\omega^2$. This one fixed witness works for every allowed family matrix. Hence all choices have a negative radial direction in the stated truncation. $\square$

Equation (22) is independently checked by direct finite differences of V_F plus its counterterm, and Eq. (23) is saturated by three equal masses $M_i = \mu_0 e^{-1/4}$. These checks do not replace the analytical global bound. The result rules out “let a later family fit repair the current hard radial block”; it does not rule out the Spin(10) action. A controlled scalar/IR reorganization or a new consistently defined scalar point must be considered before a physical fit.

5 A computed linear- C_{HN} finite threshold subset

5.1 Direct vertex integral and terminal matching

Use the existing no-half convention

$$\mathcal{L} \supset -\frac{1}{2}N^T M N - Y_{\alpha i}(L_{\alpha} H)N_i + C_{ij}N_i N_j(H^{\dagger} H) + \text{h.c.}, \quad M(\rho) = M - 2C\rho, \quad (26)$$

where $C = C_{HN}$, $\rho = H^{\dagger} H$ and $C^T = C$. The Weinberg convention is $C_{5,\text{tree}}^I = -YM^{-1}Y^T$. This is the convention used by the existing RG implementation and its 2024 dimension-five checks [3]. Work at leading $q_H/M_i^2 = 0$ and linear order in C .

In a positive Takagi basis, the Yukawa- C bubble contains a chirality-flipping propagator $M_i/(k^2 - M_i^2)$ and a massless scalar propagator. The no-half $CNN\rho$ vertex supplies a factor two. The renormalized scalar integral at zero external momenta is

$$b_i = -\int_0^1 \log \frac{tM_i^2}{\mu^2} dt = 1 + \log \frac{\mu^2}{M_i^2}. \quad (27)$$

With the displayed interaction signs, the proper Yukawa correction is

$$\Delta Y = \frac{2}{\mathcal{L}} Y \text{diag}(M_i b_i) C. \quad (28)$$

At this order a one- C Higgs tadpole changes its mass but not its kinetic metric; the corresponding massless Higgs tadpole on a sterile line is scaleless. This does not discard other canonical type-I wave functions, which belong to different spurion contributions.

When all sterile fields are removed, the linear- C contribution from this vertex is

$$\Delta C_5^{(C)} = -\Delta Y M^{-1} Y^T - Y M^{-1} \Delta Y^T. \quad (29)$$

Differentiating the same fermion determinant at $M(\rho) = M - 2C\rho$ gives the finite Higgs mass contribution

$$\Delta q_H^{(C)} = \frac{4}{\mathcal{L}} \sum_i M_i^3 \text{Re } C_{ii} \left(\log \frac{M_i^2}{\mu^2} - 1 \right). \quad (30)$$

Here q_H denotes the coefficient of $+\rho$ in the potential, in the repository's reference units. Equations (28)–(30) are a calculated subset, not the full C_{HN} seesaw matching map.

Their scale derivatives reproduce the independently implemented C -linear beta differences. For example,

$$\frac{d\Delta C_5^{(C)}}{d \log \mu} = -\frac{4}{\mathcal{L}} (Y M C M^{-1} Y^T + Y M^{-1} C M Y^T). \quad (31)$$

The masses in this expression are diagonal positive Takagi masses. The implementation tests generic complex family covariance, degenerate $O(3)$ invariance, an independent Feynman-parameter integral and direct fermion-determinant differentiation. Finite constants are supplied by the integral, not inferred solely from a beta function.

5.2 Tree elimination generates a further operator

Partition the Takagi basis into removed h and retained r blocks. Although $M_{hr} = 0$ at $\rho = 0$, generally $C_{hr} \neq 0$. Exact algebraic elimination at nonzero ρ gives

$$Y_r^{\text{eff}}(\rho) = Y_r - Y_h(M_h - 2C_{hh}\rho)^{-1}(-2C_{hr}\rho) \quad (32)$$

$$= Y_r + D_6\rho + O(\rho^2), \quad D_6 = 2Y_h M_h^{-1} C_{hr}. \quad (33)$$

In the present sign convention the Lagrangian therefore contains $-D_6(LH)N_r(H^\dagger H) + \text{h.c.}$. This is a known class of dimension-six sterile-neutrino operators [4]; its coefficient here is derived from the existing action, not added as a new free matrix. The same expansion gives

$$M_r^{\text{eff}}(\rho) = M_r - 2C_{rr}\rho - 4C_{rh}M_h^{-1}C_{hr}\rho^2 + O(\rho^3), \quad (34)$$

$$C_5^{\text{eff}}(\rho) = C_5 - Y_h M_h^{-1} Y_h^T - 2Y_h M_h^{-1} C_{hh} M_h^{-1} Y_h^T \rho + O(\rho^2). \quad (35)$$

Only Eq. (33) is needed for the particular linear- C mixed threshold consistency check below; the other generated higher operators have not been declared irrelevant for full matching.

5.3 A precise mixed-block RG residual

Let $\mathcal{C} = C_5 - Y_r M_r^{-1} Y_r^T$ denote the active-seesaw composite below the threshold. In Eq. (28) restrict the internal massive line to the removed block. This determines the hard ΔY_r and the heavy-heavy contribution to ΔC_5 . Their contribution to $\Delta \mathcal{C}$ is

$$\Delta \mathcal{C}_h = \Delta C_{5,hh} - \Delta Y_r M_r^{-1} Y_r^T - Y_r M_r^{-1} \Delta Y_r^T. \quad (36)$$

Subtracting the existing dimension-five-only beta functions leaves the identity

$$\frac{d\Delta \mathcal{C}_h}{d \log \mu} - (\beta_C^r - \beta_C^{h+r})_C = \frac{2}{\mathcal{L}} (D_6 M_r Y_r^T + Y_r M_r D_6^T). \quad (37)$$

One can see its origin without fitting coefficients: the omitted terms in Eq. (31) have an internal retained index and an external removed index. Replacing their removed propagator by M_h^{-1} gives the tree vertex in Eq. (33). A retained N_r -Higgs bubble with one D_6 and one Y_r insertion contributes

$$\beta_{C_5}|_{D_6 M_r} = \frac{2}{\mathcal{L}} (D_6 M_r Y_r^T + Y_r M_r D_6^T) \quad (38)$$

in this convention, closing precisely this residual. The finite retained-line bubble is an EFT graph to subtract, not a missing hard term to add a second time to C_5 .

This is not a claim that every dimension-six contribution must always be kept in a leading dimension-five calculation. If M_r/M_h is treated as a further small expansion parameter, the displayed effect can be consistently higher order. However, a finite matching calculation that keeps the retained masses without this extra hierarchy truncation needs the generated operator or an explicit bound on what was dropped. No such bound has yet been established for a physical fitted P54 trajectory.

The actual scalar-tree boundary can have $C \propto M$, but the existing RG equations do not preserve that alignment for generic Y_ν . Reading the old two synthetic-family trajectories at their live thresholds gives $\|D_6\|_{F\omega^2}$ of

$$(1.1505 \times 10^{-4}, 7.9314 \times 10^{-5}, 0), \quad (1.1899 \times 10^{-4}, 1.5040 \times 10^{-4}, 0). \quad (39)$$

The last entry vanishes because no sterile field remains. These are operator-closure diagnostics, not fitted observables. The old trajectories were inspected without applying the incomplete new matcher. Full dimension-six running, nonzero- q_H terms, correlated finite M_R , λ , wave-function maps and the remaining C_5 /mixed/box diagrams are still missing.

6 Updated acceptance gates and reproducibility

The common-renormalization verifier passes 46/46 checks; the C_{HN} closure verifier passes 20/20. The strengthened canonical fit interface passes 18/18 and the existing upper-vector subset remains 38/38. Passing tests includes correctly detecting a failed physical stability condition; it is not a claim that all physical gates passed.

The shared fit consumer now additionally requires common parameter/tadpole consistency, whole-background stability, mass-operator control and C_{HN} operator closure. Its previous doublet and diagram flags alone are insufficient. The frozen point is not promoted to a fit. No full Wilson/box or lower gauge/ghost/Yukawa completion is claimed.

The immediate research choices are:

1. Keep the action and first test a consistently organized scalar background with the required Goldstone/IR and momentum/gauge treatment. The theorem above specifies exactly which previous truncation cannot be rescued by a family fit; it does not substitute for this further analysis.
2. If a controlled background cannot be obtained at this scalar point, reselect common scalar parameters subject to stationarity and full radial and nonradial stability, then recompute the common spectrum/matching. Changing only the finite counterterm or fitting an isolated mass formula is not a consistent alternative. No such parameter change is made here.
3. For sequential seesaw, choose and document either the enlarged operator basis with retained-mass feedback or a justified hierarchy expansion and remainder bound. Then finish the finite diagrams and lower matching before the complete light-state and flavor fit.

New artifacts, relative to `route_f/`, are `code/verify_p54_common_renormalization.py` and `code/verify_p54_chn_threshold_closure.py`, with matching JSON/Markdown files under `output/`. The first takes `--cache-dir` pointing to the old `tmp/p54_full_doublet_cw` cache. Only three new mixed polynomial Hessians were needed, stored in the worktree cache; all later runs reuse them. Reports record current source hashes, matrices, individual checks and explicit non-completion flags.

References

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